
Keras: Deep Learning Simplified

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01

Introduction to Keras

Core problem: complexity vs speed



Deep learning complexity

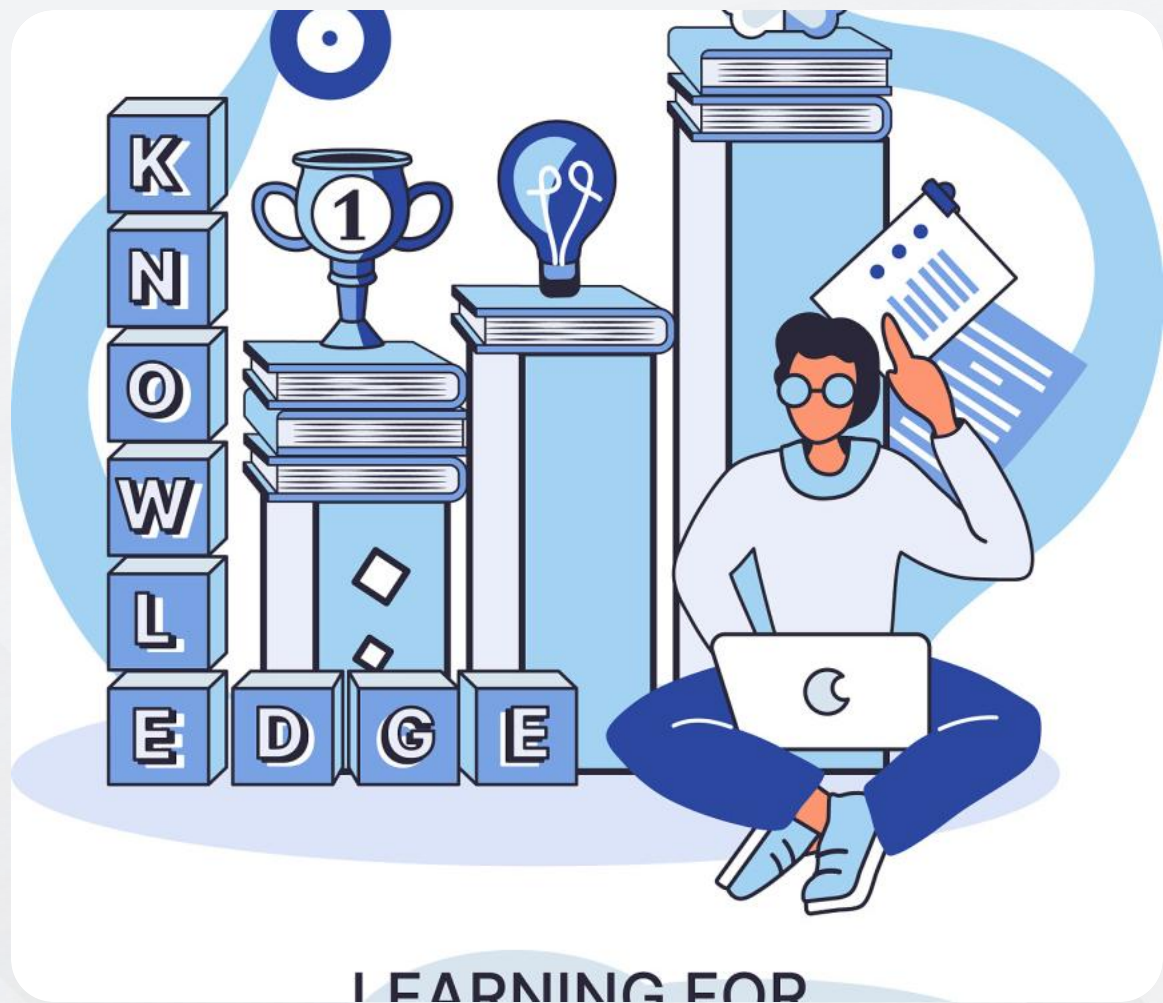
The challenge of managing complexity in deep learning development.



Development speed importance

The need for rapid iteration to keep up with project demands.

Defining Keras: high-level API



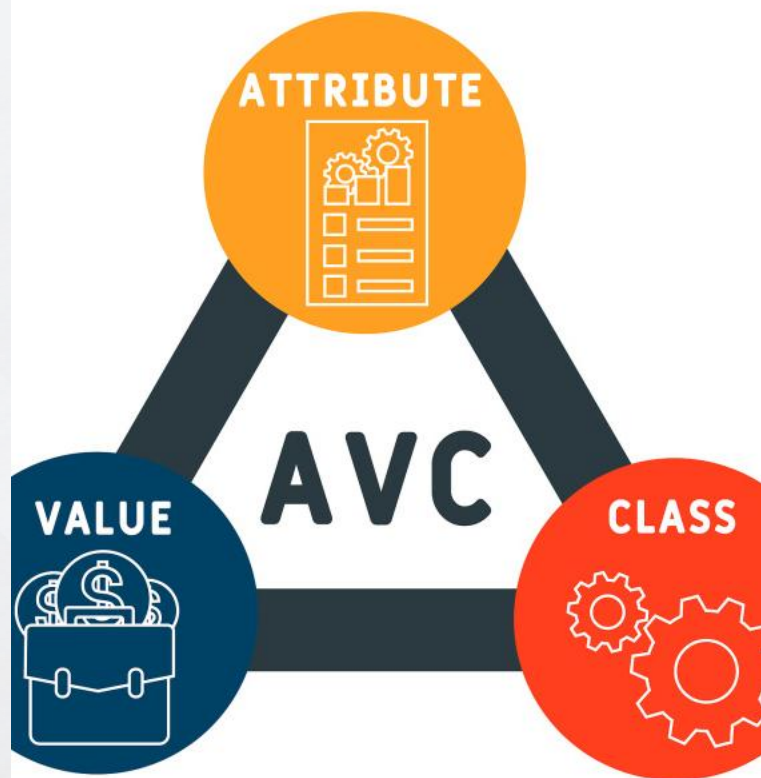
Official high-level API

Keras is the official high-level API for deep learning.

Designed for fast iteration

Keras is specifically built to enable quick prototyping and iteration.

Strategic value to saascv



Rapid prototyping of CV models

Keras allows for swift development of Computer Vision models.

Enabling saascv's innovation

Keras provides the strategic advantage of accelerating saascv's model development.



02

Architecture of Simplicity

Sequential API for basic models

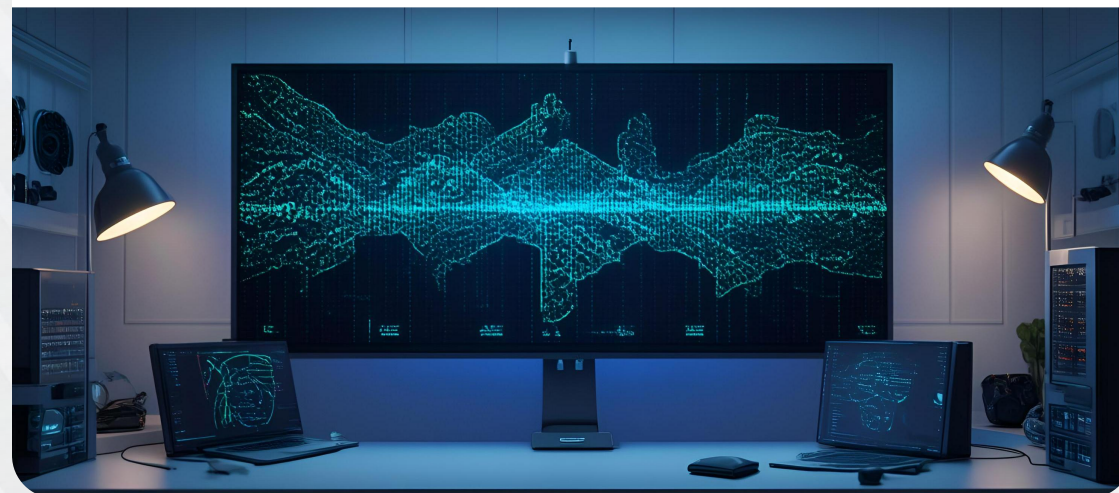


Linear stack of layers

Sequential API allows for simple, linear model creation with a stack of layers.

Suitable for basic models

Designed for straightforward, entry-level deep learning tasks.



Functional API for complex networks

01

Complex, non-linear networks

Functional API handles complex network architectures with multiple inputs/outputs.

02

Shared layers support

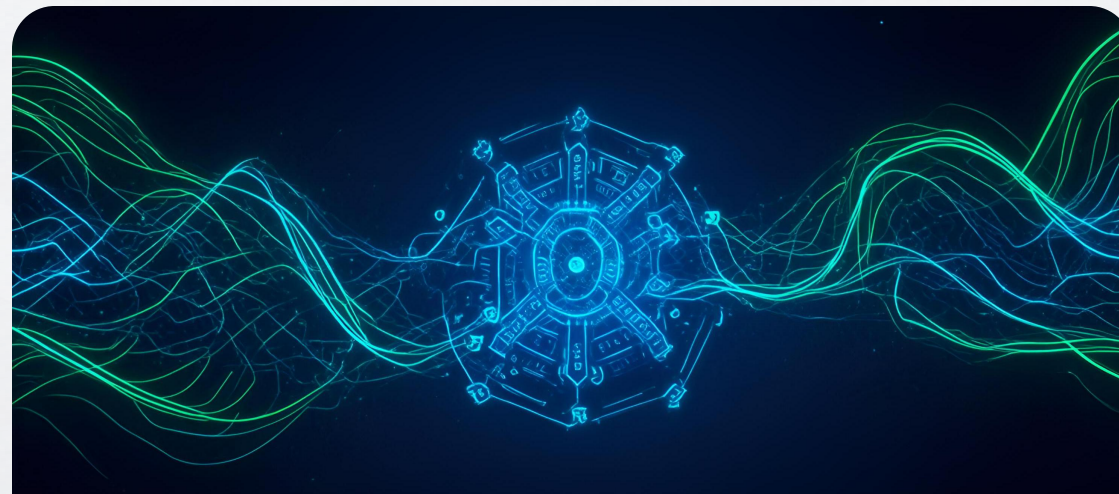
Enables the creation of models with shared layers, essential for complex structures.

Model Subclassing for custom research



Custom research workflows

Model Subclassing provides flexibility for advanced research and unique model designs.



Out-of-box workflows

Allows for the creation of models that don't fit into the Sequential or Functional paradigms.

Keras Workflow overview

Define data

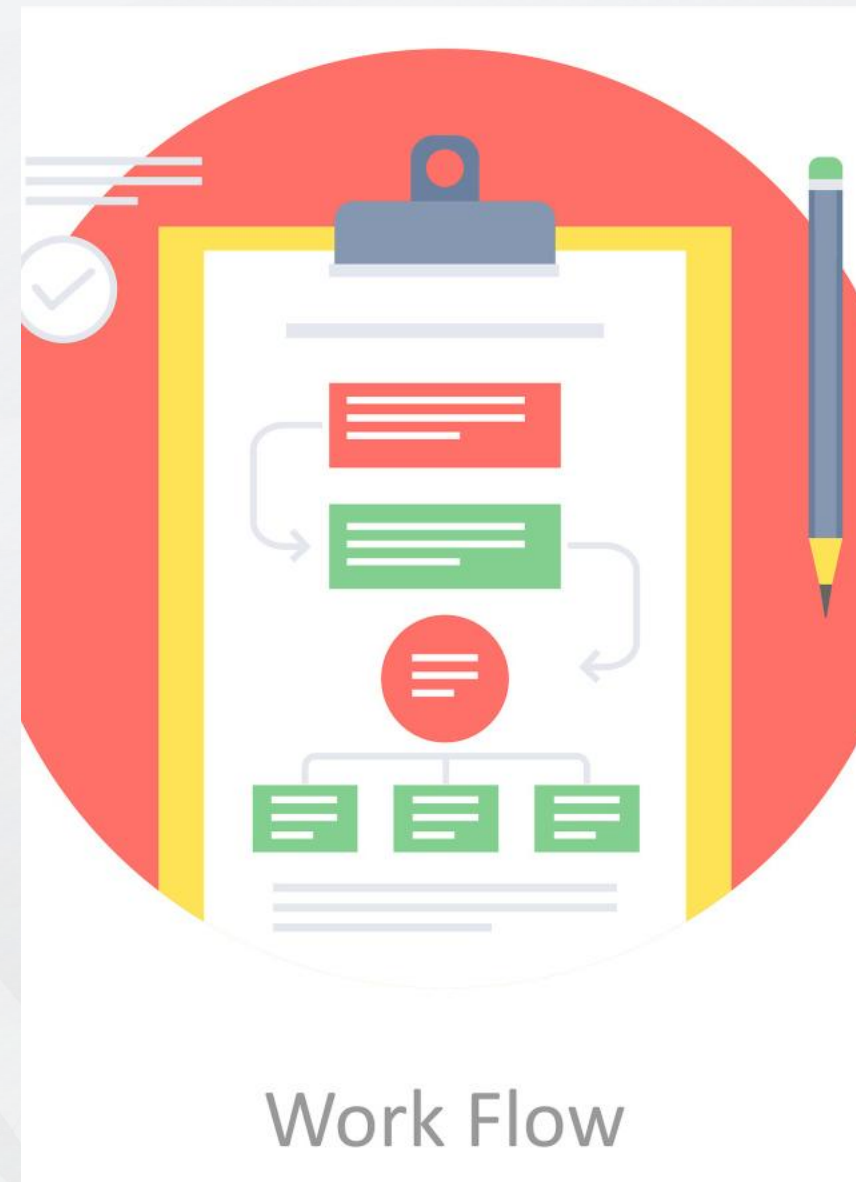
The first step in the Keras workflow is to define the input data.

Configure learning process

Includes setting up the learning rate, optimizer, and loss function.

Iterate for improvement

Iteratively train and refine the model based on performance metrics.





03

Keras Toolkit for Computer Vision

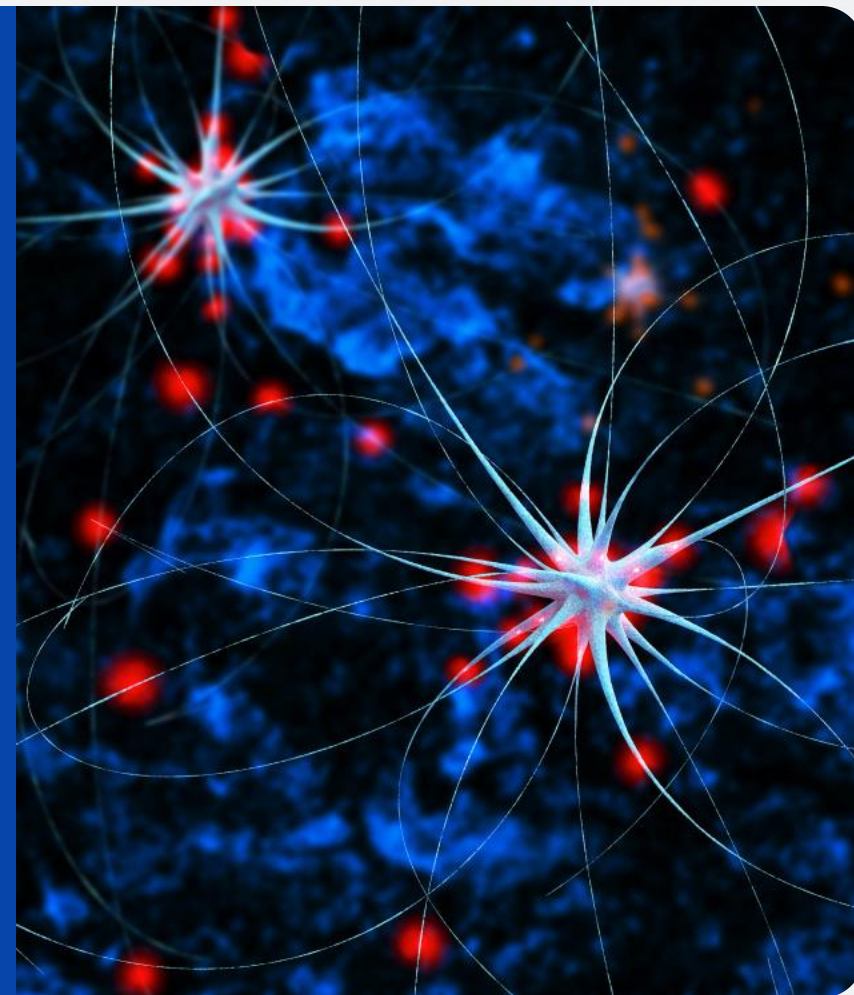
Foundational Layers: Conv2D, MaxPooling

Convolutional Layers (Conv2D)

Conv2D layers extract features from image data, essential for computer vision tasks.

Pooling Layers (MaxPooling)

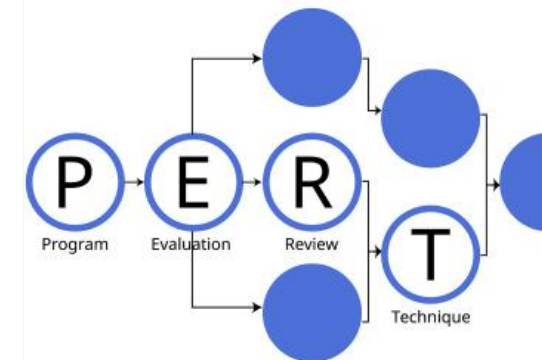
MaxPooling reduces dimensionality, retaining important features while discarding irrelevant information.



Transfer Learning with pre-trained models

Pre-trained models (VGG, ResNet)

Utilize established models like VGG and ResNet to accelerate development and leverage existing knowledge.



Fine-tuning for specific needs

Adapt pre-trained models by fine-tuning the classification head to meet saascv's unique requirements.



**Learning to Use
New Technology**

Data Augmentation techniques



Artificial dataset expansion

Use techniques like RandomFlip and RandomRotation to increase dataset size and model robustness.



Improving model robustness

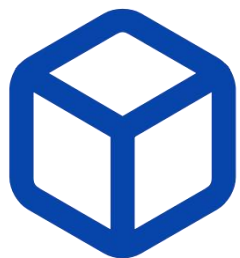
Data augmentation helps models generalize better by simulating diverse training conditions.



04

Keras 3 Revolution

Shift to multi-backend API



Keras as multi-backend API

Keras now supports TensorFlow, PyTorch, JAX, and more.



Adaptability to different backends

Allows leveraging the strengths of each backend for specific tasks.

Strategic advantage of code portability



Code portability in Keras

Enables seamless switching between different deep learning frameworks.



Utilizing best backend for the job

Choose JAX for speed or TensorFlow for production serving based on needs.



05

Deployment and Professionalism

Callbacks for monitoring and stopping

Monitoring with TensorBoard

Professional tool for real-time tracking of model performance metrics

Early stopping mechanism

Automatically halts training when a monitored metric has stopped improving





Deployment Pipeline: TFLite, TensorFlow Serving

Exporting models for mobile/edge

TFLite enables deployment on mobile and edge devices for real-time applications

Serving models in the cloud

TensorFlow Serving facilitates the deployment of models in cloud environments

Conclusion: Keras as saascv standard

Fast, maintainable code

Keras is the standard for creating deep learning models quickly and efficiently

Enabling rapid prototyping

Keras accelerates the development of Computer Vision models for saascv projects

THE END

Thank You